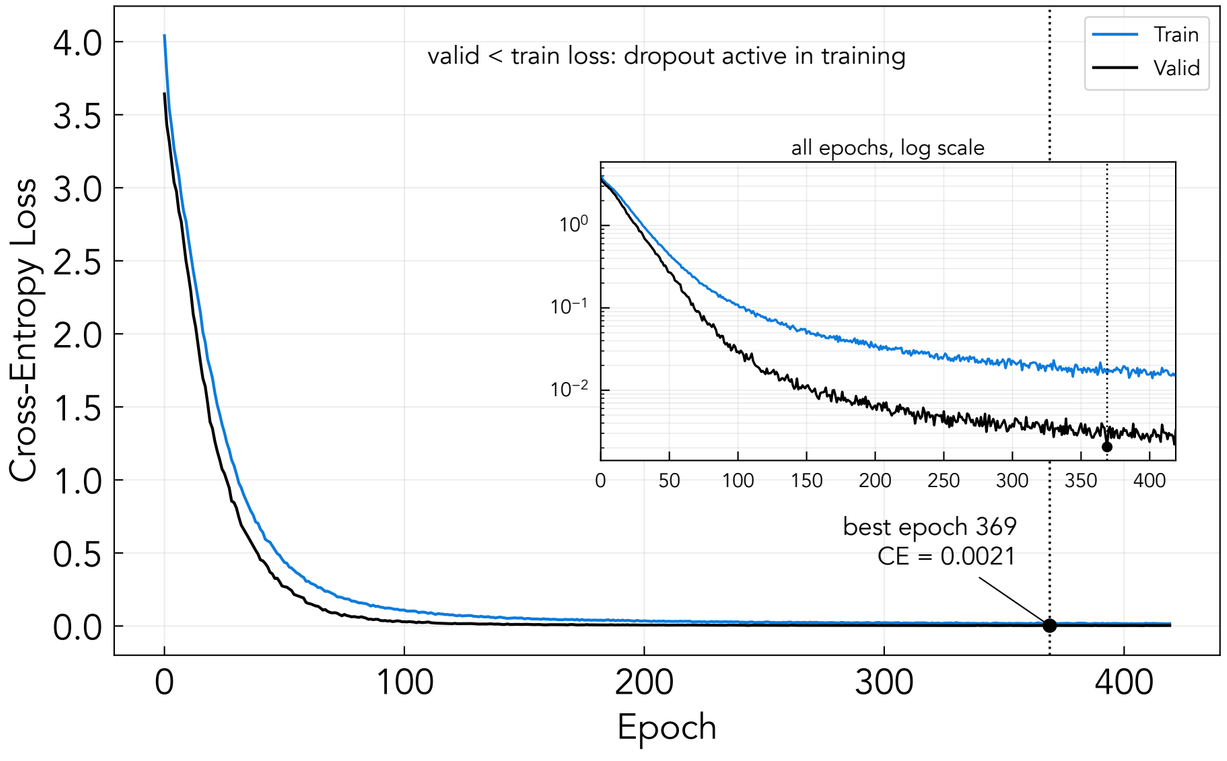
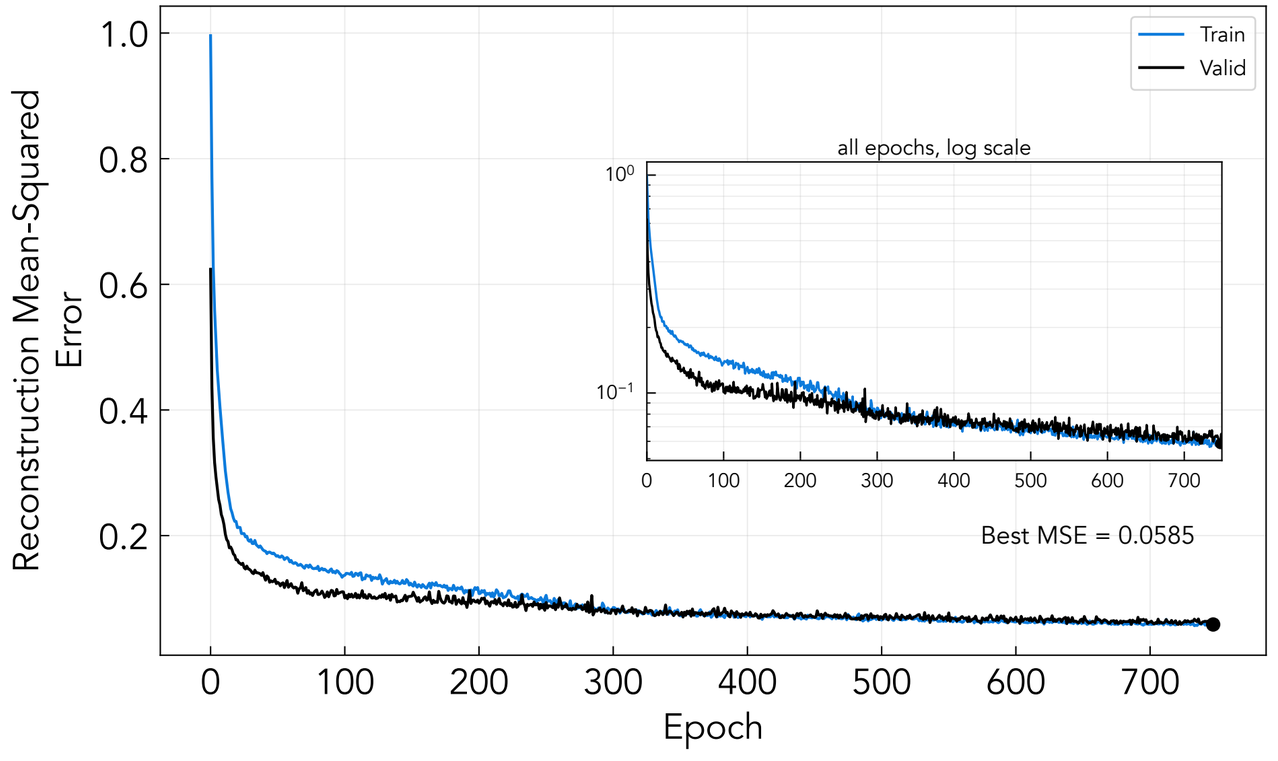
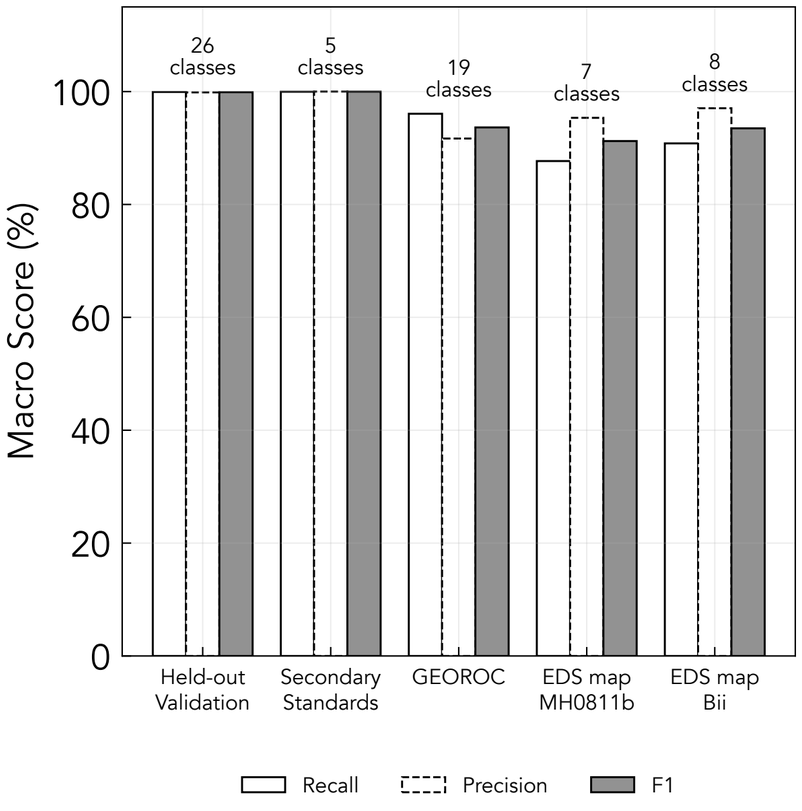
A.


Classification Loss
Train
Valid
valid < train loss: dropout active in training
all epochs, log scale
best epoch 369
CE = 0.0021
Cross-Entropy Loss
Epoch

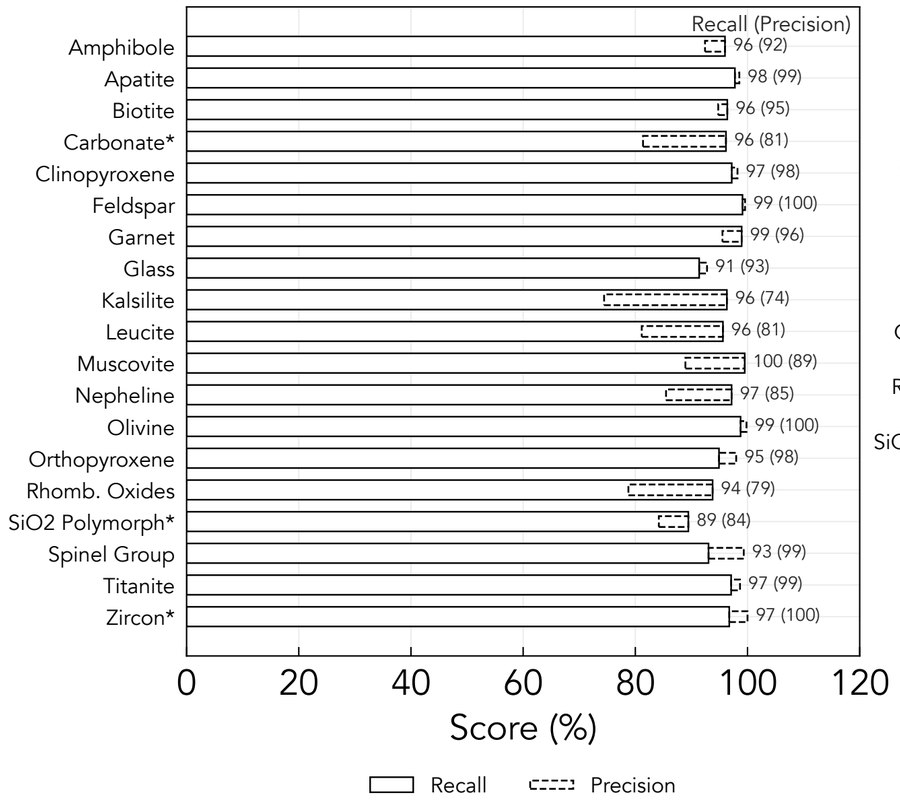
B.


Reconstruction Loss from z_2
Train
Valid
all epochs, log scale
Best MSE = 0.0585
Reconstruction Mean-Squared Error
Epoch

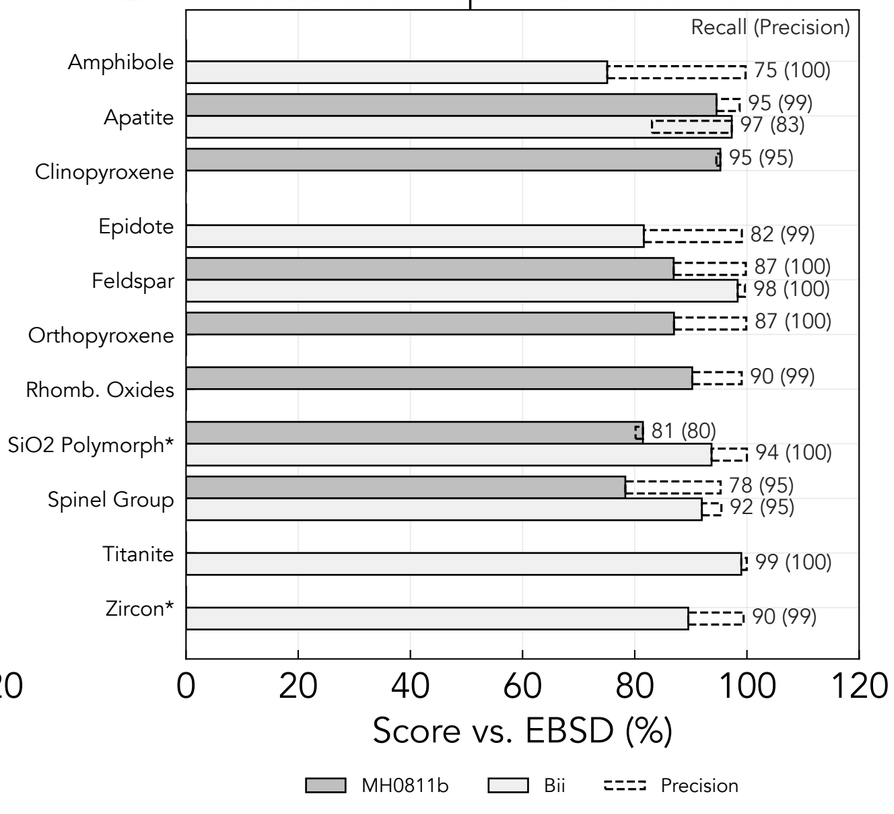
C.


Validation Dataset Metrics
26 classes
5 classes
19 classes
7 classes
8 classes
Held-out Validation
Secondary Standards
GEOROC
EDS map MH0811b
EDS map Bii
Macro Score (%)
Recall
Precision
F1

D.


GEOROC - Per-Class Metrics
Recall (Precision)
Amphibole 96 (92)
Apatite 98 (99)
Biotite 96 (95)
Carbonate* 96 (81)
Clinopyroxene 97 (98)
Feldspar 99 (100)
Garnet 99 (96)
Glass 91 (93)
Kalsilite 96 (74)
Leucite 96 (81)
Muscovite 100 (89)
Nepheline 97 (85)
Olivine 99 (100)
Orthopyroxene 95 (98)
Rhomb. Oxides 94 (79)
SiO2 Polymorph* 89 (84)
Spinel Group 93 (99)
Titanite 97 (99)
Zircon* 97 (100)
Score (%)
Recall
Precision

E.


EBSD-EDS maps - Per-Class Metrics
Recall (Precision)
Amphibole 75 (100)
Apatite 95 (99)
97 (83)
Clinopyroxene 95 (95)
Epidote 82 (99)
Feldspar 87 (100)
98 (100)
Orthopyroxene 87 (100)
Rhomb. Oxides 90 (99)
SiO2 Polymorph* 81 (80)
94 (100)
Spinel Group 78 (95)
92 (95)
Titanite 99 (100)
Zircon* 90 (99)
Score vs. EBSD (%)
MH0811b
Bii
Precision